Searching for Tau Core signal candidates in SPARC residual structure

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Abstract

Paper 1 found that externally reviewed structural disturbance is associated with larger low-acceleration residual scatter in SPARC. Paper 2 reversed the question and showed that fixed residual-shape features can recover those A/C labels better than chance, while remaining explicitly non-unique with respect to MOND-simple and empirical RAR-like baselines. This Paper 3 seed asks a narrower follow-up question: where do TPG/projection residuals, MOND-simple residuals, RAR-like residuals, and measured rotation curves point to candidate residual structure that could plausibly carry the missing observer- and environment-dependent Tau Core weight? The current packet identifies candidate galaxies, radial onset categories, and environment/observability stress channels. It does not validate Tau Core, does not numerically test RMOND, and does not claim gravity-model selection.

1 Purpose and claim boundary

The working hypothesis is that the TPG/projection baseline already carries part of the local Tau Core weighting, while the remaining TPG residual may carry missing observer- and environment-dependent weights. This is a candidate-search statement, not a proof. A defensible Paper 3 must therefore separate three layers:

1. an operational residual layer, based on frozen TPG/projection, MOND-simple, and RAR-like residual maps;
2. a systematics layer, based on distance, inclination, point count, H I kinematics, beam smearing, and non-circular motions;
3. a theory layer, where a Tau Core interpretation is allowed only if the residual pattern survives the systematics layer and is more specific than ordinary RAR/MOND behavior.

The permitted claim is that this packet defines reproducible Tau Core signal candidates. The forbidden claim is that the candidates already prove Tau Core.

2 Data inherited from Papers 1 and 2

The packet inherits derived SPARC residual artifacts from the Paper 1 and Paper 2 public packages. Raw SPARC rotmod files and raw H I survey products are not redistributed. The retained derived tables include pointwise absolute residuals for TPG/projection, MOND-simple, and empirical RAR-like families; galaxy-level residual features; distance, observability, and environment summaries; and small radial-pilot control summaries.

The main residual quantity is

$$r_{m,gi} = \left|\log V_{\mathrm{obs},gi} - \log V_{m,gi}\right|,$$

with galaxy-level burden

$$\mathrm{RMS}_{m,g} = \sqrt{\frac{1}{N_g}\sum_i r_{m,gi}^2}.$$

The TPG-specific excess used for candidate triage is

$$\Delta_{\mathrm{TPG-low},g} = \mathrm{RMS}_{\mathrm{TPG},g} - \min\left(\mathrm{RMS}_{\mathrm{MOND},g}, \mathrm{RMS}_{\mathrm{RAR},g}\right).$$

Positive values mark galaxies where the TPG/projection residual is larger than the best available low-acceleration comparator in this packet. Negative values mark TPG success controls.

3 Theory motivation

The relevant external baseline is the SPARC radial acceleration relation [4, 2] and the broader MOND/RAR literature [5, 3]. These works are not optional background; they define the strongest ordinary low-acceleration competitor. If Tau Core only reproduces the same residual ordering as RAR/MOND, it is not yet distinguished.

The systematics literature is equally central. Non-circular motions in H I velocity fields [8, 7] and beam-smearing/rotation-curve quality effects [1] can generate residual structure without new physics. The JWST/NIRCam Zone-of-Avoidance result [6] is used only as an observer/line-of-sight motivation: foreground obscuration and hidden structure can materially affect what an observer can map. It is not evidence for Tau Core by itself.

4 Candidate construction

For each galaxy we record TPG/projection RMS, MOND-simple RMS, RAR-like RMS, TPG-specific excess, environment proxy, distance, reconstruction-risk proxy, inclination, and the first radial bin where the TPG/projection absolute residual exceeds 0.15 dex. Candidate classes are assigned by frozen screening rules:

- *clean tau candidate*: high TPG residual, high TPG-specific excess, and non-high reconstruction risk;
- *environment tau candidate*: high residual burden with high environment cue;
- *TPG divergence follow-up*: positive TPG-specific excess without enough cleanliness for the first two labels;
- *TPG success control*: low residual burden where TPG is no worse than MOND/RAR-like baselines.

These classes are triage labels. They are not physical classifications.

Table 1: Candidate shortlist from the regenerated seed packet.

Galaxy	Class	TPG RMS	TPG excess	Env.	Onset	Candidate class
UGC00731	C	0.340	0.093	0.00	inner_R<0.33Rmax	clean tau
CamB	A	0.964	-0.080	1.72	inner_R<0.33Rmax	environment
UGC05764	A	0.329	0.076	0.00	inner_R<0.33Rmax	TPG divergence
NGC2403	A	0.085	0.029	1.86	outer_R>=0.67Rmax	TPG divergence
NGC3741	C	0.204	0.103	-0.12	inner_R<0.33Rmax	TPG divergence
UGC08490	C	0.186	0.055	0.00	inner_R<0.33Rmax	TPG divergence
NGC3109	C	0.176	0.039	0.20	middle_0.33-0.67Rmax	TPG divergence
UGC06446	C	0.194	0.038	0.00	inner_R<0.33Rmax	TPG divergence

5 Residual and environment readouts

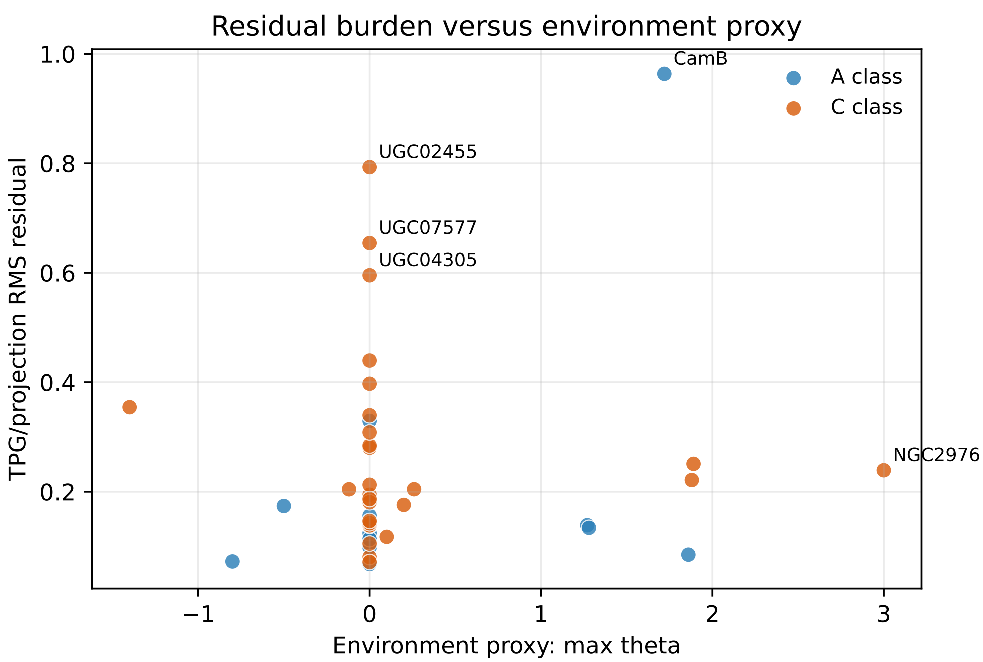

Figure 1: TPG/projection residual burden versus the inherited environment proxy. Named points are the largest seed-candidate scores.

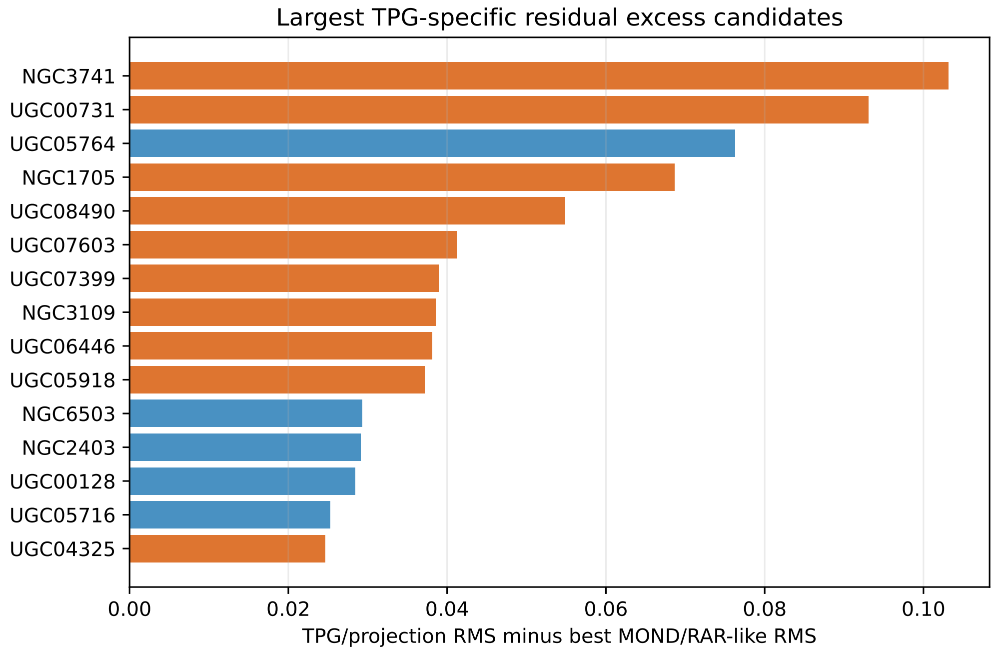

Figure 2: Galaxies with the largest TPG/projection residual excess relative to the better of MOND-simple and RAR-like RMS.

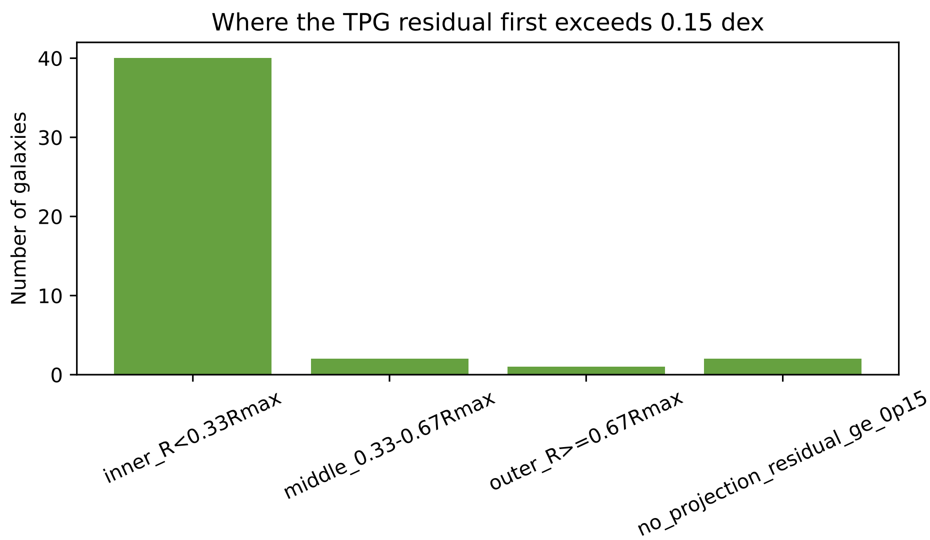

Figure 3: Radial bin where the TPG/projection absolute residual first exceeds 0.15 dex. This is a morphology-of-failure diagnostic, not a detection statistic.

Table 2: Screening correlations for TPG/projection residual burden.

Covariate	Pearson r	Interpretation
DistanceMpc	-0.291	screening_correlation_not_causal_attribution
EnvMaxTheta	0.110	screening_correlation_not_causal_attribution
W_tau_eff_abs_v01	0.983	screening_correlation_not_causal_attribution
ReconstructionRiskChannel_v01	0.317	screening_correlation_not_causal_attribution
MeanErrVobsKms	-0.214	screening_correlation_not_causal_attribution
InclinationErrorDeg	0.214	screening_correlation_not_causal_attribution

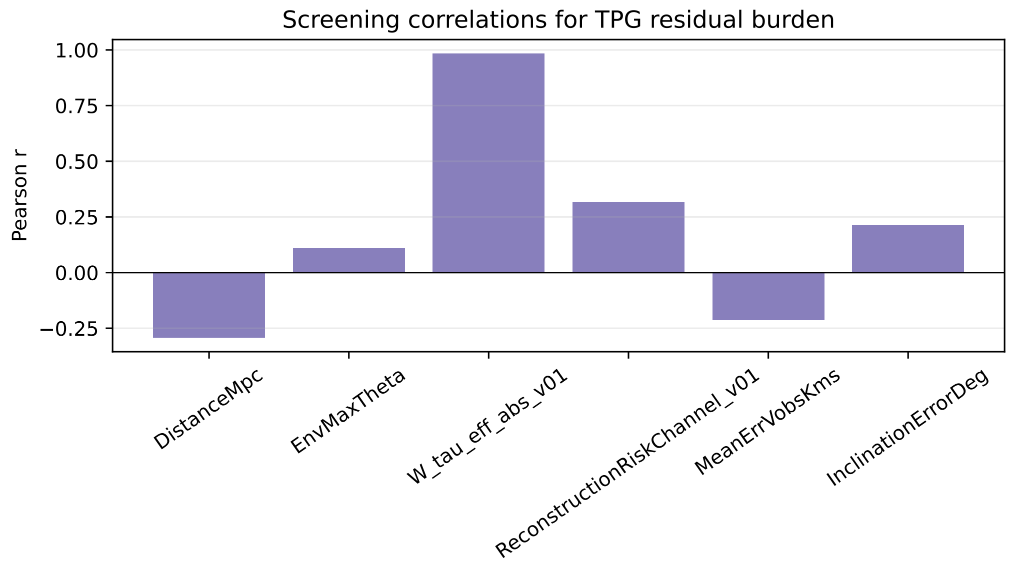

Figure 4: Screening correlations with the TPG/projection residual burden. These are not causal estimates.

6 RMOND and comparator status

The requested RMOND comparison is not yet a numeric endpoint in this seed packet. The repository contains theory notes relating TPG and RMOND, but it does not yet contain a frozen pointwise RMOND prediction/residual table on the same SPARC radii. Treating RMOND as tested without that table would weaken the paper. The next technical gate is therefore explicit: generate or import a pointwise RMOND residual map and add it to the same comparator table as TPG/projection, MOND-simple, and RAR-like baselines.

7 Interpretation

The most useful interpretation is modest. TPG success controls show where the local projection baseline is already adequate. TPG divergence follow-ups show where the residual has structure that might encode missing observer/environment weights, but these are exactly the cases where ordinary systematics can also enter. A candidate becomes interesting only when three facts hold together: the TPG residual diverges in a structured radial way, the divergence is not shared by all low-acceleration baselines, and the object has a plausible Tau Core weighting channel that is not reducible to distance, inclination, beam smearing, or non-circular motion.

8 Conclusion

This seed opens Paper 3 as a reproducible candidate-search project. It carries forward the discipline learned from Papers 1 and 2: freeze the endpoint, compare against MOND/RAR baselines, name the systematics, and avoid promoting a diagnostic pattern into a physical proof. The current evidence is enough to define targets for Tau Core follow-up, especially TPG-divergence galaxies and TPG-success controls, but it is not enough to claim a detected Tau Core field.

The next paper-grade step is a frozen RMOND residual table plus a held-out environment/line-of-sight validation rule. Only after that can the paper ask whether the residual candidates are truly Tau Core-specific rather than ordinary low-acceleration or H I-systematics behavior.

References

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